

1. Create a database called “CricketerStatsDB”.

SQL Command
Create Schema CricketerStatsDB; use cricketerstatsdb;

2. Create a table titled “Players” with the following constraints.

Field	Type	Constraints
player_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
player_name	VARCHAR (100)	NOT NULL
country	VARCHAR (50)	NOT NULL
date_of_birth	DATE	NOT NULL
batting_style	VARCHAR (20)	NOT NULL
bowling_style	VARCHAR (50)	NULLABLE

SQL Command
create table Players(player_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT, player_name VARCHAR (100) NOT NULL, country VARCHAR (50) NOT NULL, date_of_birth DATE NOT NULL, batting_style VARCHAR (20) NOT NULL, bowling_style VARCHAR (50) NULL);

3. Create a table titled “Matches” with the following constraints.

Field	Type	Constraints
match_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
match_date	DATE	NOT NULL
venue	VARCHAR (100)	NOT NULL
match_type	VARCHAR (20)	NOT NULL

SQL Command
create table Matches(match_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT, match_date DATE NOT NULL, venue VARCHAR (100) NOT NULL, match_type VARCHAR(20) NOT NULL);

4. Create a table titled “Performances” with the following constraints.

Field	Type	Constraints
performance_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
player_id	INT	NOT NULL
match_id	INT	NOT NULL
runs_scored	INT	NOT NULL DEFAULT 0
balls_faced	INT	NOT NULL DEFAULT 0
wickets_taken	INT	NOT NULL DEFAULT 0
overs_bowled	DECIMAL (4,1)	NOT NULL DEFAULT 0.0

SQL Command
<pre>create table Performances(performance_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT, player_id INT NOT NULL, match_id INT NOT NULL, runs_scored INT NOT NULL DEFAULT 0, balls_faced INT NOT NULL DEFAULT 0, wickets_taken INT NOT NULL DEFAULT 0, overs_bowled DECIMAL (4,1) NOT NULL DEFAULT 0);</pre>

5. Populate the table “Players” with the following data

player_name	country	date_of_birth	batting_style	bowling_style
Virat Kohli	India	1988-11-05	Right-Handed	NULL
Joe Root	England	1990-12-30	Right-Handed	Off Spin
David Warner	Australia	1986-10-27	Left-Handed	NULL
Shaheen Afridi	Pakistan	2000-04-06	Left-Handed	Left-arm Fast
Kane Williamson	New Zealand	1990-08-08	Right-Handed	Off Spin

SQL Command
<pre>INSERT INTO Players (player_name, country, date_of_birth, batting_style, bowling_style) VALUES ('Virat Kohli', 'India', '1988-11-05', 'Right-Handed', NULL), ('Joe Root', 'England', '1990-12-30', 'Right-Handed', 'Off Spin'), ('David Warner', 'Australia', '1986-10-27', 'Left-Handed', NULL), ('Shaheen Afridi', 'Pakistan', '2000-04-06', 'Left-Handed', 'Left-arm Fast'), ('Kane Williamson', 'New Zealand', '1990-08-08', 'Right-Handed', 'Off Spin');</pre>

6. Populate the table “Matches” with the following data.

match_date	venue	match_type
2024-03-15	MCG, Australia	Test
2024-04-02	Lord's, England	ODI
2024-04-18	Eden Gardens, India	T20
2024-05-01	Gaddafi Stadium, Pakistan	Test

SQL Command
INSERT INTO Matches (match_date, venue, match_type) VALUES ('2024-03-15', 'MCG, Australia', 'Test'), ('2024-04-02', 'Lord"s, England', 'ODI'), ('2024-04-18', 'Eden Gardens, India', 'T20'), ('2024-05-01', 'Gaddafi Stadium, Pakistan', 'Test');

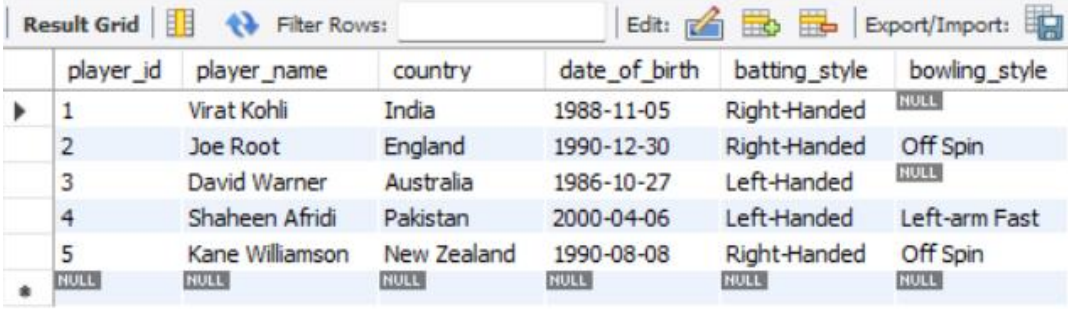
7. Populate the table “Performances” with the following data.

player_id	match_id	runs_scored	balls_faced	wickets_taken	overs_bowled
1	1	120	200	0	0.0
2	1	150	250	0	5.0
3	2	80	70	0	0.0
4	4	15	10	5	20.0
5	3	45	30	0	2.0
1	3	60	45	0	0.0
2	4	55	50	0	4.0

SQL Command
INSERT INTO Performances (player_id, match_id, runs_scored, balls_faced, wickets_taken, overs_bowled) VALUES (1, 1, 120, 200, 0, 0.0), (2, 1, 150, 250, 0, 5.0), (3, 2, 80, 70, 0, 0.0), (4, 4, 15, 10, 5, 20.0), (5, 3, 45, 30, 0, 2.0), (1, 3, 60, 45, 0, 0.0), (2, 4, 55, 50, 0, 4.0);

Query-01:

Display all players.

SQL Command						
select * from players						
Output:						
						

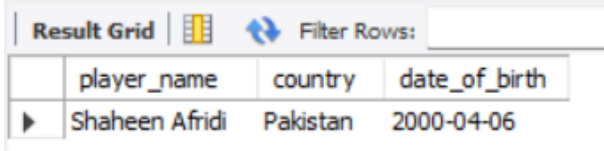
Query-02:

List all match venues and types.

SQL Command		
select venue, match_type from matches		
Output:		
		

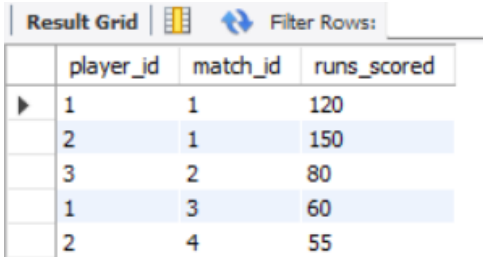
Query-03:

Find players born after 1990.

SQL Command			
select player_name, country, date_of_birth from players where year(date_of_birth)>1990;			
Output:			
			

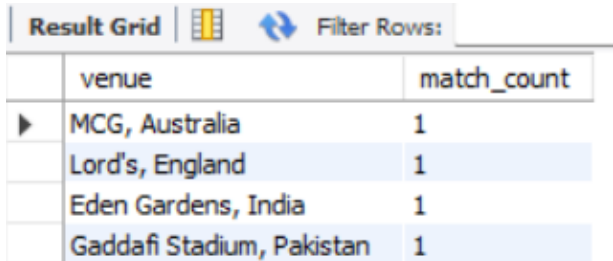
Query-04:

Show performances where runs scored > 50.

SQL Command			
select player_id,match_id,runs_scored from performances where runs_scored>50;			
Output:			
			

Query-05:

Find the number of matches played at each venue.

SQL Command		
select venue, count(venue) as match_count from matches group by venue;		
Output:		
		

Query-06:

List players with their total runs scored.

SQL Command	
select player_name, sum(runs_scored) as total_runs from players natural join performances group by player_id	

Output:

	player_name	total_runs
▶	Virat Kohli	180
	Joe Root	205
	David Warner	80
	Shaheen Afridi	15
	Kane Williamson	45

Query-07:

Find players who scored more than 100 in any match.

SQL Command

```
select player_name, runs_scored
from players as p natural join performances as pf
where pf.runs_scored>100 and (p.player_id=pf.player_id);
```

Output:

	player_name	runs_scored
▶	Virat Kohli	120
	Joe Root	150

Query-08:

Find average balls faced per player.

SQL Command

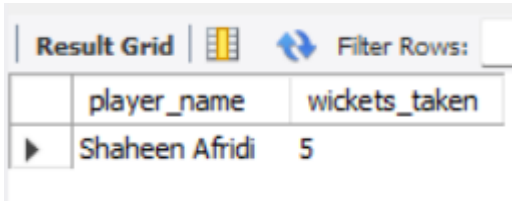
```
select player_name, avg(balls_faced) as avg_balls
from players as p natural join performances as pf
group by player_id
```

Output:

	player_name	avg_balls
▶	Virat Kohli	122.5000
	Joe Root	150.0000
	David Warner	70.0000
	Shaheen Afridi	10.0000
	Kane Williamson	30.0000

Query-09:

Find the top wicket-taker player.

SQL Command	
<pre>select player_name, wickets_taken from players natural join performances where wickets_taken= (select max(wickets_taken) from performances);</pre>	
Output:	
	

Query-10:

Find the top wicket-taker player.

SQL Command	
<pre>select match_id,venue, sum(runs_scored) as total_runs, sum(wickets_taken) as total_wickets from matches natural join performances group by match_id order by match_id asc;</pre>	
Output:	
